

PROGRAM-4: Integrate Git with Docker for source controlled Application builds

Project Structure:

Dockerfile

[index.js](#)

package.json

package-lock.json

Step 1: Create a package.json file using command `npm init -y`

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ npm init -y
Wrote to /home/1rv24mc017_ananya_ganapati_hegde/DevOps/lab/p4/package.json:
```

```
{
  "name": "p4",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^5.1.0"
  },
  "devDependencies": {},
  "repository": {
    "type": "git",
    "url": "git+https://github.com/ananyahegde/DevOps-P4.git"
  },
  "bugs": {
    "url": "https://github.com/ananyahegde/DevOps-P4/issues"
  },
  "homepage": "https://github.com/ananyahegde/DevOps-P4#readme",
  "description": ""
}
```

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$
```

Step 2: Create a [index.js](#) file

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ nano server.js
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ cat server.js
const express = require('express');
const app = expressA();
const PORT = 3000;

app.get('/', (req, res) => {
    res.send("Hello from docker with github integration!");
})

app.listen(PORT, () => {
    console.log("Listening at http://localhost:${PORT}");
})
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ █
```

Step 3: Add the dependencies

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ npm install express
removed 3 packages, changed 5 packages, and audited 66 packages in 4s

19 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ █
```

Step 4: Create a Dockerfile

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ nano Dockerfile
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ cat Dockerfile
FROM node:18-slim AS build

WORKDIR app/
COPY package*.json ./
RUN npm install
COPY . .

FROM node:18-slim

WORKDIR app/
COPY --from=build app/ app/
EXPOSE 3000

CMD ["node", "server.js"]

1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$
```

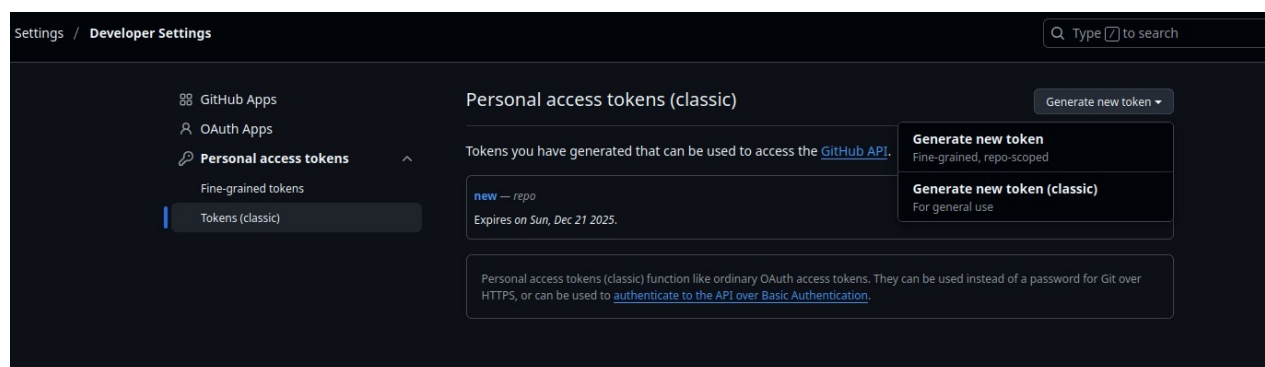
Step 5: Login to your github account and Create a new Repository

Login→ New→ New Repository→ Provide the repository name

Step 7:Generate the personal access token

Go to profile->Settings->Developer Settings->Personal Access Token->Token(classic)->Generate Token

Save it somewhere since it is visible for some time only



Step 8: Initialize the git repository locally

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ ls -al
total 52
drwxrwxr-x  3 1rv24mc017_ananya_ganapati_hegde 1rv24mc017_ananya_ganapati_hegde 4096 Nov 21 10:06 .
drwxrwxr-x  7 1rv24mc017_ananya_ganapati_hegde 1rv24mc017_ananya_ganapati_hegde 4096 Nov 21 10:03 ..
-rw-rw-r--  1 1rv24mc017_ananya_ganapati_hegde 1rv24mc017_ananya_ganapati_hegde  193 Nov 21 10:01 Dockerfile
drwxrwxr-x 67 1rv24mc017_ananya_ganapati_hegde 1rv24mc017_ananya_ganapati_hegde 4096 Nov 21 09:34 node_modules
-rw-rw-r--  1 1rv24mc017_ananya_ganapati_hegde 1rv24mc017_ananya_ganapati_hegde  265 Nov 21 09:34 package.json
-rw-rw-r--  1 1rv24mc017_ananya_ganapati_hegde 1rv24mc017_ananya_ganapati_hegde 28135 Nov 21 09:34 package-lock.json
-rw-rw-r--  1 1rv24mc017_ananya_ganapati_hegde 1rv24mc017_ananya_ganapati_hegde  253 Nov 21 09:36 server.js
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1rv24mc017_ananya_ganapati_hegde/DevOps/lab/p4/.git/
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ git add .
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ git commit -m "inital commit"
[master (root-commit) 9de7466] inital commit
 636 files changed, 72754 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 node_modules/.package-lock.json
 create mode 100644 node_modules/accepts/HISTORY.md
 create mode 100644 node_modules/accepts/LICENSE
 create mode 100644 node_modules/accepts/README.md
 create mode 100644 node_modules/accepts/index.js
 create mode 100644 node_modules/body-parser/HISTORY.md
 create mode 100644 node_modules/body-parser/LICENSE
 create mode 100644 node_modules/body-parser/README.md
 create mode 100644 node_modules/body-parser/index.js
 create mode 100644 node_modules/body-parser/lib/read.js
 create mode 100644 node_modules/body-parser/lib/types/json.js
 create mode 100644 node_modules/body-parser/lib/types/raw.js
 create mode 100644 node_modules/body-parser/lib/types/text.js
 create mode 100644 node_modules/body-parser/lib/types/urlencoded.js
 create mode 100644 node_modules/body-parser/lib/utils.js
 create mode 100644 node_modules/body-parser/package.json
 create mode 100644 node_modules/bytes/History.md
 create mode 100644 node_modules/bytes/LICENSE
```

Step 9: Connect to the remote repository and push the code to the repository

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ git remote add origin git@github.com:ananyahegde/DevOps-P4.git
error: remote origin already exists.
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ git branch -M main
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$ git push -u origin main
Enumerating objects: 719, done.
Counting objects: 100% (719/719), done.
Delta compression using up to 12 threads
Compressing objects: 100% (666/666), done.
Writing objects: 100% (719/719), 725.34 KiB | 1.70 MiB/s, done.
Total 719 (delta 142), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (142/142), done.
To github.com:ananyahegde/DevOps-P4.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p4$
```

Output:

ananyahedge / DevOps-P4

Go to file

1 Branch

0 Tags

Go to file

Add file

<> Code

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

DevOps-P4

Public

Pin

Watch

Fork

Star

main

1 Branch

0 Tags

Go to file

Add file

<> Code

ananyahedge initial commit

0b4e91c · 21 minutes ago

1 Commit

node_modules	initial commit	21 minutes ago
Dockerfile	initial commit	21 minutes ago
package-lock.json	initial commit	21 minutes ago
package.json	initial commit	21 minutes ago
server.js	initial commit	21 minutes ago

README

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About

DevOps Lab Program 4 Demo

Activity

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Releases

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Create a new release

Packages

No packages published

Publish your first package

Suggested workflows

Based on your tech stack

Publish Docker Container

Configure

Build, test and push Docker image to GitHub Packages.

Grunt

Configure

Build a NodeJS project with npm and grunt.

Deno

Configure

Test your Deno project

More workflows

Dismiss suggestions